BONUS POINTS - POKÉMON DAY III: 25 Points Possible Halloween Arcades

10/28/2026

Attempt 1

In Progress

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10/17/2026 to 10/28/2026

Details

POKÉMON DAY III: HALLOWEEN ARCADES

An Interactive Analytics Simulation and Mini-Game Development Challenge

Course: Analytics Application
Coverage: Chapters 1–6, including Prescriptive Analytics
Participants: 10 groups, consisting of 4–5 members each
Exhibition Date: October 27, 2026 (Morning and Afternoon)
Bonus Points: Minimum of +25 points toward the Prelims Examination

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I. THE CHALLENGE

Welcome to **Pokémon Day III: Halloween Arcades!**

Throughout our discussions, we have established one important principle: **Analytics can be applied to everything.**

Now, it is time to put that principle into practice.

For this activity, each group will conceptualize, develop, and operate its own **Pokémon-inspired, Halloween-themed mini-game or simulation system**, which will serve as a playable arcade booth on October 27, 2026.

The challenge is to build an interactive system where players can make decisions, generate data, experience different outcomes, and use analytics to improve their strategies.

Unlike conventional arcade games, your systems must go beyond entertainment. They must demonstrate how data can be collected, analyzed, interpreted, and transformed into meaningful

predictions and recommendations.

By the end of the activity, we should have **10 different arcade booths**, each demonstrating a unique and creative application of Analytics.

II. COMPLETE CREATIVE FREEDOM

There will be **NO assigned game concepts, genres, scenarios, or prescribed technologies**.

Each group has complete creative freedom in designing its arcade.

You may create a strategy game, management simulation, adventure, survival challenge, puzzle, competition, resource-allocation game, or any other concept you can imagine.

You are free to decide:

- The game's concept, storyline, setting, mechanics, and objectives.
- The Pokémon elements and Halloween theme.
- The real-world or fictional problem being simulated.
- Whether the game is competitive, cooperative, multiplayer, or based on individual sessions.
- The type of data to be used or generated.
- The software, programming languages, frameworks, and technologies.
- The analytics techniques, algorithms, and visualizations.
- The way analytics will interact with and influence gameplay.

Pokémon-related data, characteristics, mechanics, and other relevant information may be incorporated into the game's design and analytical environment.

The only nonnegotiable academic condition is that your system must meaningfully apply the principles and concepts discussed throughout Chapters 1–6.

Creativity is encouraged, but the application of analytics must be genuine, technically defensible, and relevant to the game's objectives.

III. INTEGRATION OF CHAPTERS 1–6

Each mini-game must demonstrate the complete analytics journey, from identifying a problem to recommending a decision.

Chapter	Expected Application

1. Introduction to Analytics Application	Demonstrate how analytics adds value to the game and improves decisions or outcomes.
2. Framing Analytics Problems	Establish the problem, objectives, measurable KPIs, decision variables, and constraints.
3. Data for Modern Analytics	Obtain or generate meaningful data, prepare it, manage it, and make it available for analysis.
4. Exploratory & Diagnostic Analytics	Present performance summaries, trends, comparisons, and evidence-based explanations of outcomes.
5. Predictive Analytics	Apply an appropriate predictive technique to anticipate relevant future results, risks, probabilities, or performance.
6. Prescriptive Analytics	Evaluate possible actions and recommend appropriate strategies based on data, objectives, and constraints.

The Four Analytics Questions

Your system must meaningfully answer the following:

1. **Descriptive Analytics — What happened?**
   Show what players accomplished, what decisions were made, and what the data reveal.
2. **Diagnostic Analytics — Why might it have happened?**
   Investigate patterns, differences, relationships, or contributing factors associated with the outcomes.
3. **Predictive Analytics — What is likely to happen next?**
   Use an appropriate predictive technique to estimate a future outcome, probability, or performance level.
4. **Prescriptive Analytics — What should we do?**
   Recommend a feasible strategy or course of action based on the available evidence, objectives, and constraints.

These components must be integrated into the game's functionality, not simply presented as separate explanations or decorative dashboards.

You are not required to use every mathematical formula or algorithm discussed in class. You are required to choose the appropriate analytical techniques for your particular problem.

Predictive models must have a justifiable evaluation method. Prescriptive recommendations must be derived from meaningful comparisons, rules, or decision models rather than arbitrary suggestions.

IV. DATA REQUIREMENTS

Each group may independently choose how its system will obtain or generate data.

Acceptable approaches include:

Data Approach	Description
Real-World Data	Existing datasets or records relevant to the game or simulation.
Synthetic Data	Artificially generated data representing defined characteristics, trends, or relationships.
Simulated Data	Data produced by computational models, game rules, scenarios, or simulated events.
Player-Generated Data	Data collected from actual gameplay, including decisions, actions, results, and interactions.
Hybrid Data	Any combination of these approaches.

There is no required preference for one approach over another.

There is also **no fixed minimum number of records**. Groups must determine and justify the amount of data needed for their selected analytical methods.

However, each system must demonstrate:

- Meaningful and organized data structures.
- Relevant data collection or generation processes.
- Appropriate data preparation and validation.
- The ability to record actual gameplay decisions or outcomes.
- Data sufficient to support the intended analyses.
- Clear connections between the data, analytics, and recommendations.

You may begin with existing or generated data and allow the dataset to grow as visitors interact with your system.

V. THREE-DEVICE ARCADE REQUIREMENT

Each group must operate its arcade using **at least three (3) separate physical devices**.

These devices must function as components of a common interactive system.

Recommended Setup

Device	Recommended Function
Device 1 — Arcade Control Center	Game monitoring, data management, analytics, visualizations, predictions, and recommendations.

Device 2 — Player Terminal A	Actual gameplay, decisions, interactions, and data generation.
Device 3 — Player Terminal B	Gameplay for another player, generating additional information for the same system.

Groups are free to create a different device configuration as long as all three devices actively contribute to the game or simulation.

Technical Requirements

1. At least two devices must allow visitors to participate in actual gameplay.
2. The system must accommodate simultaneous or overlapping player sessions.
3. Player interactions must contribute to a common dataset or coordinated simulation state.
4. The analytics component must process or reflect data generated by gameplay.
5. The three devices must communicate or synchronize through an appropriate technical architecture.
6. The system must support repeated gameplay sessions and updated analytical results.

You may use laptops, tablets, smartphones, or combinations of these devices.

The system may operate through a local network, shared server, web application, or another workable setup.

Simply displaying the same static website or presentation on three devices does not satisfy this requirement.

Your group must demonstrate a genuinely functional multi-device arcade.

VI. MINIMUM SYSTEM FEATURES

A. Playable Mini-Game

Your system must include:

- An interactive interface.
- Clear gameplay objectives, rules, and mechanics.
- Meaningful player decisions.
- Measurable results or performance indicators.
- Different possible outcomes.

- The ability to replay or start new sessions.

As much as possible, a visitor should be able to complete a meaningful round in approximately **3–5 minutes**.

B. Functional Analytics

Your game must generate or process data and apply the four categories of analytics.

The analytical components may appear as dashboards, prediction panels, decision-support tools, in-game advisers, strategy comparisons, performance reports, or any other creative interface.

You are free to determine how these components will be presented.

C. Data-Driven Decisions

Your game's analytics must affect or support decisions.

For example, players may be able to compare strategies, explore alternative scenarios, inspect predictions, consider recommendations, and observe the consequences of their choices.

Your system should make analytics useful to the player rather than merely displaying calculations.

D. Technical Reliability

Your arcade must be functional during the exhibition, including:

- Reliable interaction across the required devices.
- Correct recording and retrieval of gameplay data.
- Working analytics computations.
- Appropriate handling of invalid inputs.
- The ability to begin new sessions.
- Reasonable recovery from connectivity or application interruptions.

Groups must prepare their own devices, connectivity arrangements, chargers, and other equipment necessary to operate their booth.

VII. OCTOBER 27 — THE HALLOWEEN ARCADE EXHIBITION

On **Tuesday, October 27, 2026**, the classroom will become an interactive Pokémon-themed Halloween arcade.

All 10 groups will establish and operate their respective booths.

There will be two exhibition sessions.

Morning Session

Each group must:

- Set up its arcade booth and activate all required devices.
- Accommodate visitors and allow actual gameplay.
- Demonstrate its system's functionality.
- Generate and record gameplay data.
- Explain the game's analytical components when requested.

Afternoon Session

Each group must:

- Continue operating its arcade.
- Accommodate additional gameplay sessions.
- Demonstrate the continued reliability of the system.
- Present analytics based on gameplay and available data.
- Show how predictions and recommendations support player decisions.

Groups may adjust their presentation and booth operations between sessions as needed.

Each group must organize member assignments to ensure that its arcade remains operational and properly attended.

VIII. PRELIMS EXAMINATION BONUS POINTS

To encourage system development and actual implementation, bonus points will be awarded through three separate milestones.

Milestone	Bonus Points	Qualification
Completed and Functional System	**+15 minimum**	Immediately upon successful system verification by the instructor.

October 27 — Morning Operation	**+5 minimum**	Successful operation and actual visitor participation during the morning exhibition.
October 27 — Afternoon Operation	**+5 minimum**	Successful operation and actual visitor participation during the afternoon exhibition.
Total Baseline Opportunity	**+25**	Upon successful achievement of all three milestones.

1. System Completion Bonus (+15)

Once your group finishes developing its system, you may immediately present it for verification.

Upon confirmation that the system satisfies the mandatory gameplay, analytics, and three-device requirements, **+15 points will immediately be credited to the Prelims Examination** of qualifying group members.

You do not have to wait until October 27 to obtain this bonus.

An incomplete prototype, nonfunctional system, or static demonstration will not qualify for system completion.

2. Morning Arcade Bonus (+5 Minimum)

Successfully operating your booth during the morning session of October 27 will earn **at least +5 additional points**.

The system must be available for actual visitor interaction, with the required functionality operating during the session.

3. Afternoon Arcade Bonus (+5 Minimum)

Successfully operating your booth during the afternoon session will earn **at least +5 additional points**.

The system must remain operational and continue supporting gameplay and analytics.

4. Additional Bonus Opportunities

The instructor may award bonuses above the exhibition minimums based on:

- Exceptional creativity and game design.
- Outstanding application of analytics.
- Particularly impressive technical implementation.

- Excellent gameplay and visitor engagement.
- Meaningful use of accumulated data.
- Strong explanation and defense of analytical choices.

Additional points are discretionary and not guaranteed.

All awarded bonus points will be applied to the Prelims Examination.

Bonus eligibility also depends on individual contribution and participation. Members must be able to demonstrate and explain their assigned responsibilities.

IX. SYSTEM VERIFICATION

To receive the system-completion bonus, your group must demonstrate the functioning system directly to the instructor.

The following will be checked:

Component	Verification
Gameplay	A player can start, interact with, and finish a meaningful game session.
Three-Device Functionality	Three separate devices operate as part of the same system.
Data Management	Gameplay events, decisions, and outcomes are properly recorded.
Descriptive Analytics	The system accurately summarizes relevant data and results.
Diagnostic Analytics	The system investigates factors associated with performance or outcomes.
Predictive Analytics	The system produces a justified prediction and can explain how its performance was evaluated.
Prescriptive Analytics	The system evaluates feasible choices and produces meaningful recommendations.
End-to-End Integration	The group can demonstrate the connection between gameplay, data, analysis, and decisions.
Technical Reliability	The system can support repeated sessions and maintain reliable operation.

The instructor may test the application using different inputs, gameplay conditions, or scenarios.

System verification will be conducted through an actual demonstration. No recorded demonstration video is required.

X. SUBMISSION REQUIREMENTS

Each group must submit **one Google Drive folder link through Canvas** containing the following:

1. Working System

The application files, source code, executable build, deployment link, or other materials needed to inspect and run the application.

2. Data and Analytical Resources

The relevant datasets, data structures, analytical scripts, models, workflows, or computational resources used by the system.

3. Brief System Documentation

A concise document containing:

- Game title, concept, and objectives.
- Gameplay rules and instructions.
- System architecture and three-device configuration.
- Data sources or generation methods.
- Application of Chapters 1–6.
- Predictive and prescriptive methods.
- Instructions for running the system.
- Known limitations, if any.

4. Arcade Exhibition Evidence

Relevant gameplay/session logs or other verifiable records showing that the system operated during the morning and afternoon sessions.

The instructor will also verify exhibition participation through direct observation.

No video presentation, gameplay recording, or demonstration video is required.

XI. IMPORTANT REMINDERS

1. **AI-assisted development is allowed.** You may use AI to assist with programming, interface design, game development, debugging, data generation, and analytics. However, you must understand and be capable of explaining your system.
2. **You are free to choose your development tools.** No particular programming language, framework, database, or analytics platform is required.
3. **Use the right analytical methods.** Advanced models are not automatically better. The appropriateness, correctness, and usefulness of the method matter more.

4. **Creativity matters.** The Pokémon and Halloween atmosphere should be recognizable, but the game concept, implementation, and analytical problem are yours to determine.
5. **Prioritize gameplay and visitor experience.** Your arcade must be understandable, interactive, and enjoyable.
6. **Analytics must be functional.** Static charts, arbitrary recommendations, or prewritten explanations that do not correspond to the underlying data are insufficient.
7. **All members must contribute.** Each group member must have a meaningful responsibility in the development, preparation, or operation of the arcade.
8. **Prepare for the exhibition.** Your group is responsible for providing the necessary equipment and ensuring that your system can operate throughout the required sessions.

XII. THE FINAL CHALLENGE

For Pokémon Day III, you are not simply building a Halloween arcade game.

You are building a system that demonstrates how analytics can transform decisions and outcomes.

Your players should be able to do more than finish a game and receive a score. They should experience how data can help them understand their performance, discover patterns, anticipate possibilities, and choose better strategies.

Ten groups. Ten original arcades. One challenge: Turn Data into Decisions.

Pokémon Day III: Halloween Arcades — October 27, 2026.

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